Keep the signal.
Keep the inconvenient results.

A complete, explicitly synthetic training-run monitor: 20 runs, three metrics, and a million-point stress case.

> **SHIP the generated-data frontend. DON'T SHIP it as a live training control.**
> The frontend is checked locally. Real-source ingestion, identity, freshness/revision contracts, authorization and operator acceptance remain prerequisites for real-money decisions. Publication is separate and awaits the user's hosting choice.

1 / Why the chart is truthful

A dedicated Web Worker owns numeric Float64 columns. Steps are sorted numerically; the last received duplicate wins, even if invalid. NaN and Infinity produce null gaps, never zero. Missing training steps also break the line. Per step bucket, first/last/min/max measurements are retained in their original order, independently for loss, learning rate and gradient norm. No smoothing or averaging is enabled.

The single loss spike is 9.4 at step 500,000. It survives every tested viewport containing it. Hover is explicitly labeled sampled; it displays an actual retained value. Precise step ranges and zoom expose detail. Log mode discloses excluded nonpositive samples and keeps a linear alternative. CSV is clearly a sampled viewport, including blank gap entries.

2 / Executed checks

41 unit tests + 13 production-browser scenarios passed. TypeScript, ESLint and production build passed; dependency audit: zero known vulnerabilities. Automated axe WCAG 2 A/AA checks on the inspected desktop/mobile screens: zero violations, without claiming full accessibility certification.

Unit tests exercise the million-point spike across six widths and six ranges, extrema/source-value identity, non-finite values, duplicate/order/gap rules, invalid latest values, zero handling, exact statistics, worker races/abort/recovery, numeric labels and UTC rollover. Browser tests independently inspect painted Canvas spike pixels at overview, narrow and three-point zoom; check visible NaN gaps, log warnings, quick switching, loading, retry, CSV, keyboard controls and mobile overflow.

Limits: these tests do not prove scientific validity, live-cluster correctness, every possible dataset/zoom, screen-reader usability or physical-device ergonomics. A subpixel gap may be invisible in an overview; counts disclose missingness. Extrema lines can exaggerate dense oscillations; zoom is needed for detailed interpretation.

3 / Failure history preserved

The initial browser suite passed 10 scenarios and failed mobile overflow. An absolutely positioned accessibility label escaped the table scroller; positioning that scroller fixed it. Original JSON, screenshot and trace remain in `evidence/e2e-first-run/`. A low-contrast footer, uPlot handler type error, duplicate/reorder counts and run metadata mismatches were also corrected and rechecked. Performance regressions below are retained.

4 / Performance in numbers

Chrome 156, macOS arm64, CPU throttle 4x on the page target. Controlled fixture: identical uPlot loss chart and canonical data; baseline draws 1M values on the main thread, optimized uses the actual worker/sampler. Three runs, medians shown. First render includes source preparation and two animation frames; zoom includes resampling; hover includes two animation frames.

Measurement	Raw baseline	Worker + extrema
First render	601.1 ms	243.2 ms
Zoom	6.9 ms	33.3 ms
Hover	30.6 ms	30.9 ms
Page + worker V8 / backing stores	57.1 MB	34.6 MB
Renderer RSS, approximate	526.5 MB	507.9 MB
Retained loss entries	1,000,000	3,580

Zoom regressed; hover is effectively unchanged. Worker resampling trades latency for bounded rendering and main-thread memory. Separate full React route, one observed run: render 548.3 ms, zoom 192.3 ms, hover 41.2 ms; V8/backing stores 37.0 MB; renderer RSS 549.9 MB.

The application sample includes automation transport and is not directly comparable with fixture timings. Worker slowdown equal to the page's 4x is not asserted. V8 includes measured worker backing stores; RSS sums renderers in an isolated one-page browser and can include spare-renderer overhead. GPU/browser/utility processes are excluded. Seven original traces, screenshots and exact methodology: evidence/performance/results.json. Reproduce with npm run test:perf after starting production preview.

5 / Silent failures checked

- **Staleness:** all fixtures are explicitly static/synthetic. Recorded status is not a live feed. Reload preparation time never claims source freshness. First real-source priority: revision, ingestion timestamp and missing-value contract.
- **Run-switch races:** query identity includes run/range/width; AbortSignal and request IDs discard late replies. Run identity changes remount details. Actual browser and out-of-order unit tests check isolation.
- **Locale / timezone:** numeric step axes, explicit en-US values, UTC dates with offset/rollover tests. No timestamp is inferred from a training step.
- **False continuity:** no smoothing; invalid/missing data breaks the line. Latest invalid values stay unavailable. A divergence fixture remains discoverable.
- **Logarithmic zeros:** nonpositive rendered samples are excluded with an explicit count. Linear mode retains zeros. A fully invalid interval shows an empty chart instead of a healthy zero curve.

6 / What surprised us, and AI assistance

A plausible status-based sparkline was the easiest false data claim; it was removed. Duplicate removal also initially masqueraded as reordered data. Independent consistency checks exposed both. Dense extrema sampling still compresses timing and can hide the physical width of a gap; representative real telemetry and operator review are the next meaningful checks.

Codex and Frontend Workbench assisted design, implementation and independent reviews. Accepted: uPlot, workers, extrema sampling, explicit fixture provenance and UTC. Corrected/rejected: fabricated trends, CSS-variable Canvas colors, colliding axis labels, metadata counts and retrying a stalled worker. Executed tests, browser rendering and Chrome traces verified the result; no human or real-cluster acceptance is claimed.